

4. Linearize the system at critical points

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In general, given n functions $f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_n(\mathbf{x})$, defined for vectors $\mathbf{x}$ in $\mathbb{R}^n$, the best linear approximation to this vector valued function near the point (n -vector) $\mathbf{a}$ is

$$\begin{pmatrix} f_1(\mathbf{x}) \\ f_2(\mathbf{x}) \\ \vdots \\ f_n(\mathbf{x}) \end{pmatrix} \approx \begin{pmatrix} f_1(\mathbf{a}) \\ f_2(\mathbf{a}) \\ \vdots \\ f_n(\mathbf{a}) \end{pmatrix} + \begin{pmatrix} \partial f_1/\partial x_1 & \partial f_1/\partial x_2 & \cdots & \partial f_1/\partial x_n \\ \partial f_2/\partial x_1 & \partial f_2/\partial x_2 & \cdots & \partial f_2/\partial x_n \\ \vdots & \vdots & \ddots & \vdots \\ \partial f_n/\partial x_1 & \partial f_n/\partial x_2 & \cdots & \partial f_n/\partial x_n \end{pmatrix} \bigg|_{\mathbf{a}} (\mathbf{x} - \mathbf{a}).$$

value at $\mathbf{a}$ derivative at $\mathbf{a}$

The matrix $\mathbf{J} = \begin{pmatrix} \partial f_1/\partial x_1 & \partial f_1/\partial x_2 & \cdots & \partial f_1/\partial x_n \\ \partial f_2/\partial x_1 & \partial f_2/\partial x_2 & \cdots & \partial f_2/\partial x_n \\ \vdots & \vdots & \ddots & \vdots \\ \partial f_n/\partial x_1 & \partial f_n/\partial x_2 & \cdots & \partial f_n/\partial x_n \end{pmatrix}$ is called the **Jacobian matrix**.

We can rewrite this linearization using more compact notation as

$$\mathbf{f} \approx \mathbf{f}(\mathbf{a}) + \mathbf{J}(\mathbf{a})(\mathbf{x} - \mathbf{a}), \quad \mathbf{f} = \begin{pmatrix} f_1 \\ \vdots \\ f_n \end{pmatrix}, \mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \mathbf{a} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}.$$

Note that this is exactly the same notation we had in the 2×2 case.

To determine the stability of the system at the critical points, you evaluate the Jacobian at each critical point. If all of the eigenvalues of the Jacobian at a point $\mathbf{a}$ have negative real part, the critical point is stable. If there is an eigenvalue with positive real part, the critical point is unstable. (However, if the highest real part among eigenvalues is exactly 0, one cannot determine the stability just from the linear approximation of the system.)

3x3 linearization

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In the previous video, we studied all of the critical points of this three-by-three system of nonlinear differential equations, which we derived to describe the interactions between a species of elephant, a species of rhino, and a species of antelope.

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Jacobian Eigenvalues and Abel-Ruffini theorem

Having discovered that the Julia package ``Symbolics.jl`` doesn't support finding eigenvalues I've been pointed towards the [Abel-Ruffini][1] theorem....

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